Sanskar Jain

AI Platform Engineer | Distributed Systems & AI Agents | AWS/Kubernetes | Hyderabad, India
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Professional Summary

Software Engineer with **2+ years** building distributed, **event-driven** backend systems and **self-service REST APIs** at Verisk Analytics, plus hands-on experience designing and operating a production **AI agent** (**OpenAI API**, prompt engineering, custom orchestration) for automated workflow routing. Strong systems fundamentals -- **AWS SQS/SNS/Lambda**, **Kubernetes**, idempotent high-throughput pipelines, API/SDK design, and governance-minded reliability (token revocation, retry/backoff, dead-letter queues) -- paired with daily **AI-native development** practice (**Claude Code**, **GitHub Copilot**) and self-directed study of **transformer architectures**, **embeddings**, and **RAG**.

Technical Skills

AI & Agentic Systems: OpenAI API, Prompt Engineering, Agent Orchestration & Workflow Automation, Multi-Agent Tool-Calling, LLM Integration Patterns, AI-Assisted Development (Claude Code, GitHub Copilot)
AI/ML Foundations (Self-Study): Transformer Architectures, Embeddings, Fine-Tuning, RAG, Vector Databases, Model Evaluation
Languages: C#, Java, Python, TypeScript, SQL
Distributed Systems & Batch/Event Processing: AWS SQS, AWS SNS, AWS Lambda, Event-Driven Architecture, CQRS, Idempotency, Retry/Backoff, Dead-Letter Queues, Asynchronous High-Throughput Pipelines
Cloud & Infrastructure: AWS (EKS, ECS, S3, ElastiCache, CloudWatch), Kubernetes, Docker, Terraform, CI/CD, Azure DevOps Pipelines
API & Data: REST APIs, API Versioning, Swagger/OpenAPI, Microservices, SDK Design, MongoDB, PostgreSQL, Redis, Elasticsearch, Query Optimization
Security & Governance: OAuth 2.0, OIDC, JWT, Okta, RBAC, Session/Token Revocation, Audit-Friendly Logging, Agile/Scrum, Code Review, Git

Professional Experience

Verisk Analytics Jan 2024 -- Present
Software Engineer II Hyderabad, India

- Co-designed **RulesRunner**, a stateless microservice using **CQRS** and **AWS SQS** that offloaded compliance checks from the client, cutting wait time from **3000ms to 500ms (83%)** while enabling horizontal scaling -- a self-service platform component consumed by downstream teams.
- Architected fault-tolerant **event-driven** pipelines with idempotent processing, retry/backoff strategies, and **AWS Lambda** consumers, sustaining **500+ production messages daily** with zero message loss -- high-throughput asynchronous processing analogous to batch-platform reliability requirements.
- Engineered a scalable backend for **Xactimate's** import/export of **30+ data types** using asynchronous processing, **Amazon S3** presigned URLs, and **MongoDB**, cutting transfer time by **38%** and issues by **25%**.
- Built self-service **ASP.NET Core** REST APIs serving **200K+ weekly requests** with **Swagger/OpenAPI** documentation and **URL-based versioning** across **16 v2 endpoints**, letting client teams integrate and migrate independently without platform-team intervention.
- Established **OAuth 2.0** authentication on **AWS EKS** behind **Kong API Gateway** for **6 of 15+ platform services**, then hardened session governance with dual-layer revocation (**JTI** blocklist plus server-side invalidation) that shrank the post-logout replay window from **7 days** to **under 1 second**.
- Improved API response time by **34%** through **Redis/Amazon ElastiCache** caching, connection pooling, and query optimization -- a cost/efficiency optimization pattern for high-throughput services.
- Developed an internal **AI agent** on the **OpenAI API** using prompt engineering and custom orchestration logic, auto-creating and routing **30--50 Jira tickets per week** and eliminating repetitive manual triage for the team.
- Adopted **AI-assisted development** (**Claude Code**, **GitHub Copilot**) as a daily workflow for feature implementation, debugging, and refactoring, reviewing all generated code before merge to maintain quality standards.
- Overhauled **Azure DevOps** CI/CD pipelines with staged gates and parallel jobs, cutting deployment time by **50%** and build cycles by **33%**; wrote **xUnit/NUnit** tests reaching **90% code coverage**.

Projects

Applied LLM & RAG Systems -- Self-Directed Study 2026

- Studied **transformer/foundation-model architectures**, **embeddings**, and **fine-tuning**, then built a **RAG** pipeline (embedding model + vector store + LLM) and prototyped a **multi-agent tool-calling** workflow (planning, tool invocation, memory) beyond the single-agent production system at Verisk.
- Shipped a multi-tenant **Payroll SaaS** to production (**15+ client organizations**) as sole engineer on **Spring Boot 4/Java 17**, **React 18**, and **PostgreSQL**, with a **Spring @Scheduled** batch job automating daily attendance backfill.

Certifications & Education

AWS Certified Cloud Practitioner Amazon Web Services | Oct 2024
B.Tech in Computer Engineering Vishwakarma Institute of Technology, Pune | 2020 -- 2024 | CGPA: 8.8/10